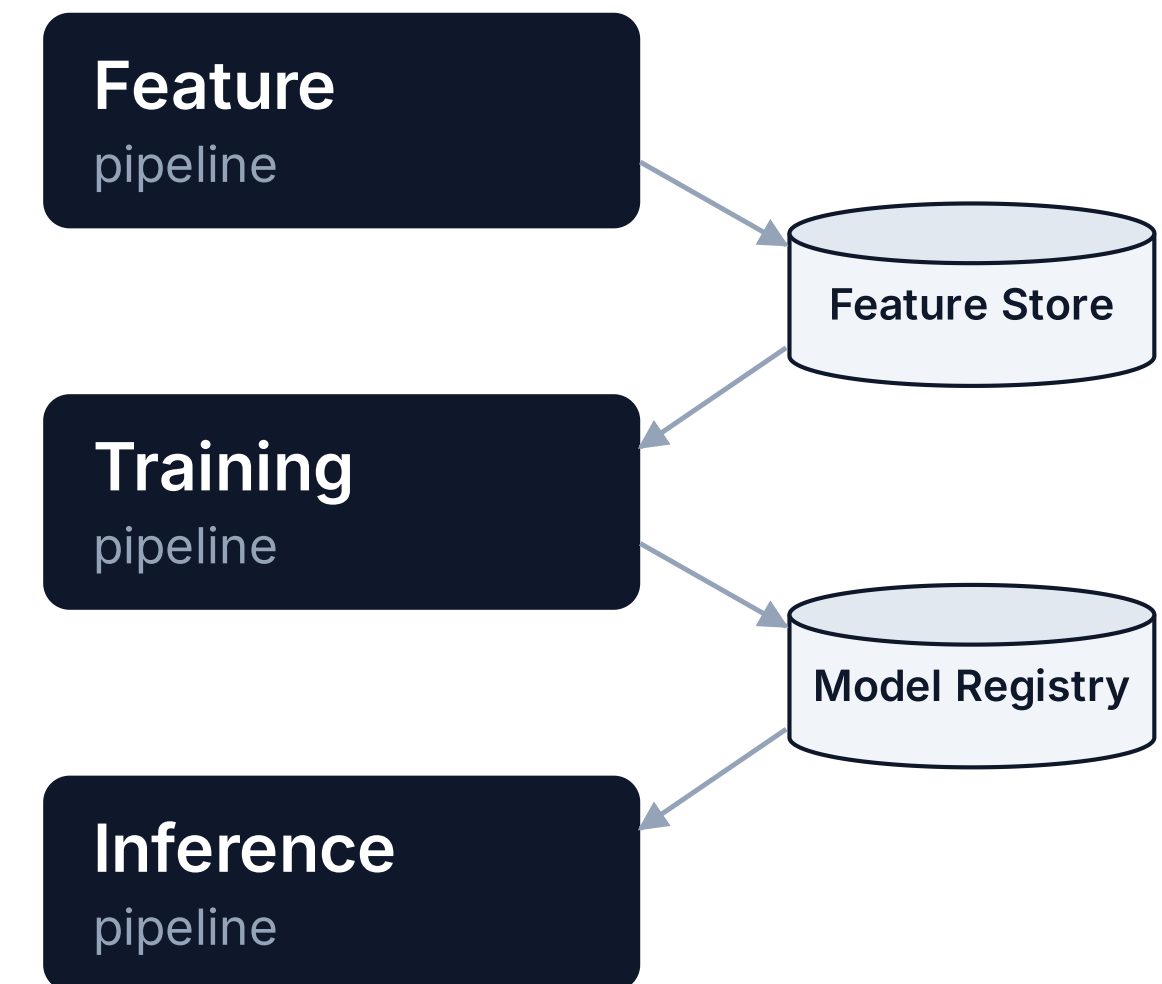


Machine Learning Operations (MLOPS)

Project · Autumn Semester 2026 (HS26)

Dr. Marc Bravin · Thu 17.09.2026



Semester Schedule

SW	DATE	TOPIC	LECTURER	MODE	PROJECT
SW01	17.09	Introduction to MLOps	Tobias, Marc	On-Site	Project Kick-Off — project + peer-review process
SW02	24.09	Batch - & Stream-Processing and FTI Architecture	Tobias	Online	—
SW03	01.10	Versioning and Containerization	Florian	On-Site	MS1 Proposal due
SW04	08.10	Container Orchestration	Florian	On-Site	MS1 peer reviews due
SW05	15.10	Data Drift and Feature Stores	Tobias	Online	—
SW06	22.10	Deployment & Serving	Marc	On-Site	—
SW07	29.10	Data Crawling	Florian	On-Site	—
SW08	05.11	LLM Inference	Josiah	On-Site	MS2 Feature Pipeline due
SW09	12.11	Efficiency Techniques	Marc	On-Site	MS2 peer reviews due
SW10	19.11	GitOps & Architectures	Florian	On-Site	—
SW11	26.11	Project: Work / Coaching	Tobias	Online	—
SW12	03.12	Project: Work / Coaching	Marc	Online	MS3 Training Pipeline due
SW13	10.12	Project: Work / Coaching	Florian	Online	MS3 peer reviews due
SW14	17.12	Final Exam (Q&A)	Marc, Florian, Tobias	On-Site	Final Exam
—	10.01	—	—	—	MS4 Live System due

Lecturers

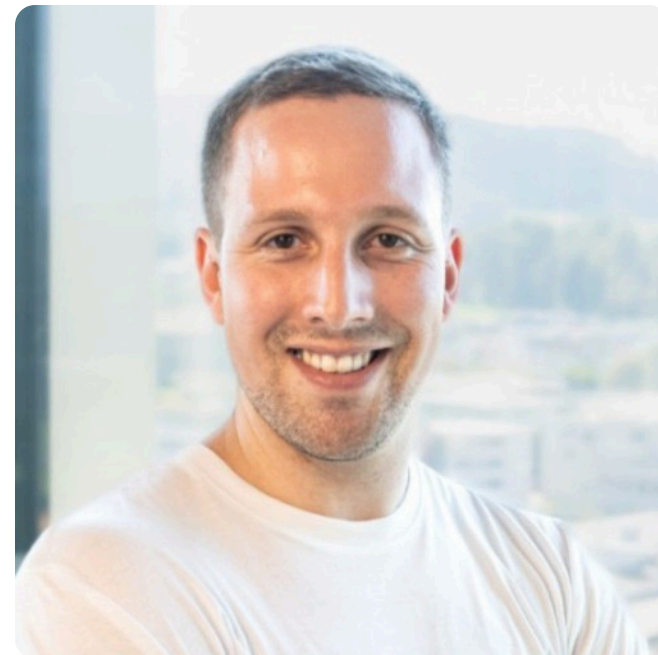
Head of the Module: Please contact Marc if you have some administrative questions!

HEAD OF THE MODULE



Dr. Marc Bravin

Lecturer at HSLU
CTO at Gopf AG



Florian Bär

Data Scientist at Helsana
Former HSLU Researcher



Tobias Mérinat

Team Lead Machine
Learning at Federal
Office for Customs and
Border Security
Former HSLU Researcher



Josiah Rohrer

Machine Learning Engineer
at Google

WHY MLOPS

Swiss AI Jobs Report 2026

+32%

more AI jobs in
Switzerland

in one year

+122%

more ML Engineer jobs
now the largest AI role

All 10

largest Swiss industries
hire more AI talent

not only IT

62%

of AI jobs are at SMEs
not only big tech

WHAT HIRING MANAGERS LOOK FOR

01

Real problems,
not textbook data

02

Depth over
prototypes

03

Concepts over
coding tests

04

Build things of
your own

05

Question the
machine

✓

Your project
counts

“Building models is no longer enough, candidates must be able to deploy and maintain them.”

HSLU AI Jobs Report 2026, on the Machine Learning Engineer role

Source: HSLU AI Jobs Report 2026, Applied AI Center. Job postings July 2025 – June 2026 vs. July 2024 – June 2025; hiring managers: “AI Job Race” interviews (pp. 32–33).

THE PROJECT

Your Semester Project

Goal: Build a “Live” ML System — an automated, cloud-ready pipeline that ingests dynamic data, computes features, trains a model, and serves predictions.

FORMAT	Individual project (single student)
DATA	No static data — dynamic sources only (crypto, weather, SBB delays...)
STACK	GitHub (public), Docker, MLflow/W&B, GitHub Actions or Airflow
DEPLOYMENT	Cloud deployment (e.g., GCP / AWS / HuggingFace) is required

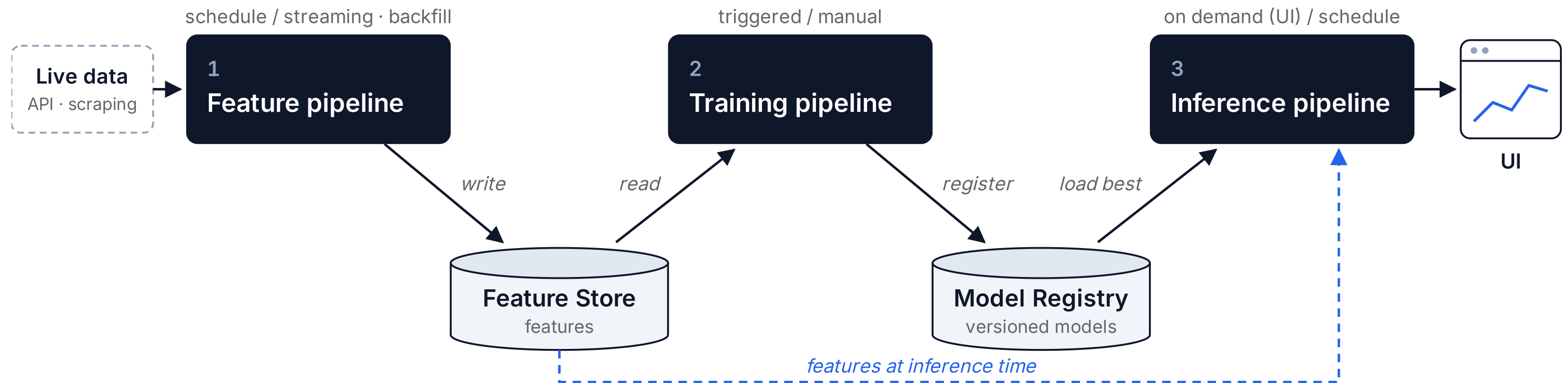
WHY NOT JUST A NOTEBOOK?

- 01 In the real world, data is dynamic and models rot over time.
- 02 “It works on my machine” is not an acceptable deployment strategy.
- 03 You are graded on the pipeline, not the accuracy.

That is why you build three automated pipelines (FTI)

THE PROJECT

The FTI Architecture



1. Feature Pipeline

Runs on a schedule or as streaming. Scrapes/fetches raw data, computes features, saves to a Feature Store. Includes backfill capability.

2. Training Pipeline

Triggered or manual. Reads from Feature Store, trains a model, evaluates it, registers to a Model Registry.

3. Inference Pipeline

Runs on demand (via UI) or on a schedule. Loads the best model from the Registry and serves predictions.

Grading

Milestone	Weight	Own Work	Peer Reviews
MS1: Project Proposal	20%	10%	10%
MS2: Feature Pipeline	20%	10%	10%
MS3: Training Pipeline	10%	5%	5%
Oral Exam	30%	30%	—
MS4: Live System (incl. Video)	20%	20%	—
Total	100%	75%	25%

SHARE OF THE MODULE GRADE



How It Works

ONE ROUND

1 Milestone due
MS1 · MS2 · MS3, 23:59

2 You write 2 reviews
one week later, 23:59

3 You receive 2 reviews
anonymous, + editorial summary

4 Response to Reviewers
in your next milestone (MS2 / MS3 / MS4)

- Your 2 projects to review are assigned at MS1 and fixed for the whole semester
- Each project receives 2 reviewers, mirroring a scientific peer review

Peer reviews = 25% of the grade,
graded on quality, not praise

YOUR REVIEW

300–500 words · anonymous · 4 sections

- 01 **Summary** — what the project does, in your own words
- 02 **Strengths** — what is well thought out
- 03 **Concerns or gaps** — what is unclear, missing or risky (be specific)
- 04 **One concrete, actionable suggestion** for the next milestone

High grade: engages with the content, names a specific technical risk, offers at least one actionable fix

Low grade: generic praise that could apply to any project ("looks good, nice idea")

RULES

- Upload both PDFs in one ILIAS "MS<n> Reviews" submission
- File name
MS<n>_P<projectID>_Reviewer<A|B>.pdf
- No name anywhere in the file, including the PDF metadata (check File → Properties)
- Late reviews are not distributed and score zero
- No reviews from you, no reviews for you: you only receive the reviews of your project if you submitted your own
- Missed your own MS1? You still owe your two reviews
- MS2 / MS3: start from the author's milestone summary, then read or run the code; a README - only review earns a low grade

What a Strong Review Looks Like

Same concern, two ways. Invented project **P07 veloforecast** (bikes per Zurich PubliBike station, 60 min ahead), MS2 review, section "Concerns or gaps".

✗ WEAK

"There are no tests anywhere in the repository. The lag features might be wrong in some cases, and error handling could be better. Add some unit tests."

could fit any repo

names no file

no evidence

"add tests" — for what?

✓ STRONG

"The 1-hour lag is positional, not temporal: `build_features.py:88` uses `shift(4)`, assuming four rows per hour. Cron skips runs: one day in your bucket has 71 of 96 snapshots, so the '1 h' lag is sometimes 75–90 min. Fix: resample each station onto a 15-min grid before shifting, and assert there are no gaps."

names the file and line

evidence: counted snapshots

actionable fix

Not longer, not cleverer: specific. Name the thing, show the evidence, say how to fix it.

Tone: respectful, objective, specific; suggest a fix that takes reasonable effort.

Responding to Reviews

AFTER EACH ROUND (MS1–MS3) YOU RECEIVE

- The 2 anonymous peer reviews of your project
- An editorial summary from us: both reviews distilled into a short list of points and decisions

IN YOUR NEXT SUBMISSION (MS2 / MS3 / MS4)

Commit docs/ms<n>_summary.pdf (max. 2 pages) with two sections:

1. **What you achieved in this milestone** (own words, key decisions, what reviewers should look at first)
2. **Response to reviews** (every point from both reviews and the editorial summary, one numbered entry each: A1, B1, E1 ..., with the commit or file)
 - Status per entry: Changed · Partly · Rejected · Not yet
 - If you disagree, say so and explain why. A reasoned rebuttal is fine; silence costs marks

EXAMPLE ENTRY A1 · No scheduling, no backfill — Changed · Added an hourly GitHub Actions cron (.github/workflows/ingest.yml) and a backfill.py for a date range; tested a 30-day backfill in CI.

Project Proposal

 **Deadline:** Thu 01.10.2026, 23:59 (ILIAS). Weight: 10% of module grade (own work)

Deliverable — max. 2-page PDF on ILIAS: four sections, graded one by one

- 01

Problem statement — what you predict, horizon, scope, success criterion with a baseline
- 02

Originality & motivation — why this problem, what's different from past projects (mlops-lab.ch, KTH ID2223)
- 03

Data source & features — live source + update frequency, the label and where it comes from, named features, leakage / split
- 04

System design — FTI diagram embedded (not ASCII), tech stack, public GitHub repo

Strong: "Predict per-station bike availability 60 min ahead, MAE ≤ 2 , beating persistence."

Weak: "Analyse bike-sharing data."

→ Also commit the proposal to your repo as docs/proposal.pdf.

Guide on ILIAS: **Proposal Guide**

Reviews · due Thu 08.10.2026, 23:59 · 10% of module grade
After the MS1 deadline you receive your 2 assigned projects privately via ILIAS. Review their proposal (docs/proposal.pdf in their repo). Upload both reviews to ILIAS "MS1 Reviews".

Feature Pipeline



Due: Thu 05.11.2026, 23:59 — GitHub repo state + summary and reviewer response file on GitHub. Weight: 10% of module grade (own work)

Deliverable — working feature pipeline:

- 01

Automated data ingestion from a live source (+ backfill)
- 02

Feature engineering & storage (feature store or versioned Parquet/GCS)
- 03

Scheduled execution (GitHub Actions / Airflow) — not manual
- 04

Docker image, pinned deps, informative README

ALSO IN THE REPO

- Summary of what you have achieved in this milestone
- Response to MS1 reviewer comments

! No secrets in the repo or its history: keep keys in `.env` (gitignored) and GitHub Secrets. [Guide on ILIAS: Repository Guide](#)

Nothing to submit: we grade your repo at the last commit before 23:59.

Reviews · due Thu 12.11.2026, 23:59 · 10% of module grade
 Review the repo of your 2 assigned projects at the MS2 deadline commit, starting from their docs/ms2_summary.pdf. Upload both reviews to ILIAS “MS2 Reviews”.

Training Pipeline

 **Due:** Thu 03.12.2026, 23:59 — GitHub repo state. Weight: 5% of module grade (own work)

Deliverable — working training pipeline:

- 01 Reads features from the MS2 feature store
- 02 Trains & evaluates a model — reproducible, no leakage
- 03 Experiment tracking with MLflow or W&B
- 04 Registers the best model to a model registry
- 05 Auto-retrain trigger (GitHub Actions / scheduled DAG)

ALSO IN THE REPO

- Summary of what you have achieved in this milestone
- Response to MS2 reviewer comments

Nothing to submit: we grade your repo at the last commit before 23:59.

Reviews · due Thu 10.12.2026, 23:59 · 5% of module grade

Review the repo of your 2 assigned projects at the MS3 deadline commit, starting from their docs/ms3_summary.pdf. Upload both reviews to ILIAS “MS3 Reviews”.

Oral Q&A

 **Date:** SW14 — Thu 17.12.2026 (in Rotkreuz). Weight: 30% of module grade

Format: individual oral exam, ~15 min per student.

No presentation.

Questions focus on your own project:

- MLOps concepts (FTI, feature store, registry, drift) in your own words
- Architecture & tool choices and the trade-offs you made
- Basis: your current repo state

LOGISTICS

planned, details follow

- Bring your laptop with your repo open
- Your exact slot and room follow by email before the exam

Just like real AI interviews: concept and architecture questions have largely replaced live coding tests (HSLU AI Jobs Report 2026).

Live System

 **Due:** 10.01.2027, 23:59 — code freeze + video pitch. Weight: 20% of module grade

Deliverables:

- GitHub repo + clone-and-run README with architecture diagram (README minimum: Repository Guide on ILIAS)
- Video pitch (max. 3 min): problem, FTI architecture, live demo on your deployed cloud URL — mandatory
- Summary of what has changed in this milestone + response to MS3 reviewer comments

Grading rubric

- 01 **Architecture completeness** — all 3 FTI pipelines + feature store
- 02 **Automation & deployment** — scheduled, auto-retrain, mandatory cloud deploy (GCP/HuggingFace)
- 03 **Inference pipeline & UI** — serving live predictions
- 04 **Observability** — tracking, deployed version known, drift/monitoring
- 05 **Code quality & reproducibility** — tests, Docker, pinned deps

Video: upload unlisted to YouTube or SWITCHtube, put the link at the top of your README and submit the same link via ILIAS “MS4 Submission”; no video files in the repo.

Showcase on mlops-lab.ch (not graded): GitHub repo description (one sentence), social-preview cover image, video link.

RULES

AI Tools, Own Work, Secrets

AI TOOLS

You may use AI tools (ChatGPT, Copilot, Claude, ...). You must understand every line of code in your repo and every decision the AI made for you. The oral exam asks exactly that: "Which code did you not write yourself, and can you explain it?" **If you can't explain it, it isn't yours.**

OWN WORK

- Individual project
- Inspiration from mlops-lab.ch and KTH ID2223 is fine; copying code or documentation is not
- We check for plagiarism

NO SECRETS

- No keys or passwords in the repo or its history
- A leaked key must be rotated; deleting the commit is not enough

Coaching and Cloud Credits

COACHING

Dedicated project work & coaching sessions



SW11 — Thu 26.11.2026 (online) — **Tobias**



SW12 — Thu 03.12.2026 (online) — **Marc**



SW13 — Thu 10.12.2026 (online) — **Florian**

Coaching is voluntary. Use these dates to work on your project! A registration sheet for the sessions follows.

GOOGLE CLOUD CREDITS · USD 50

Optional offer: USD 50 in Google Cloud credits per student. GCP is not required, but cloud deployment is; the credits are one easy way to cover it.

- 01 Request your coupon: gcp.secure.force.com
- 02 Sign in with your @stud.hslu.ch / @hslu.ch email and confirm via the email you receive
- 03 One code per email address

REQUEST & REDEEM BY

14.01.2027

VALID THROUGH

14.09.2027

FAQ

Q Can I use a pre-trained model (BERT/ResNet)?

Yes. The course is about Ops, not model architecture. Fine-tuning is fine.

Q Do I need a real feature store (Hopsworks/Feast)?

Encouraged, but versioned Parquet/CSV on GCS works as a “poor man’s feature store.”

Q What if my model performance is bad?

Graded on the pipeline, not accuracy. A fully automated pipeline at 60% is an A; a manual notebook at 99% is a fail.

Q Do peer reviews really count?

Yes — 25% of the grade, graded on quality. Generic praise earns zero.

GET STARTED

Inspiration and First Steps

Examples:

Previous HSLU MLOps projects (FS26):
mlops-lab.ch

KTH Scalable ML & Deep Learning projects:
id2223kth.github.io

- **Showcase:** we publish your projects on mlops-lab.ch. The best ones get featured on the Applied AI Center LinkedIn channel.
- **Your portfolio:** a live, public project helps you get past the “not enough experience” filter in job applications.

THIS WEEK

☐ Request your Google Cloud credits

☐ Create your public GitHub repo

☐ Pick a topic with live data

MS1 due Thu 01.10.2026